

Single core Copper & Aluminum Conductors

XLPE Insulated, Aluminum Tape armored and PVC Sheathed

Voltage Grade

12 / 20 (24) KV

Description

Soft annealed Stranded circular compacted Copper or Stranded circular compacted Aluminum conductor, semiconducting layer as conductor screen, XLPE insulated semiconducting layer as non metallic insulation screen, Copper tape or wire as metallic insulation screen, armored with aluminum tape and PVC sheathed.

Application

These cables are generally suitable for direct burial or for installation on trays or in ducts.

Technical Data

Relevant Standard	IEC 60502
Conductor	Plain copper conductor class 2 or Aluminum conductor class 2
Conductor Screen	Extruded Semiconductive Compound
Insulation	Extruded XLPE Compound Rated 90° C
Insulation Screen	Non Metallic part: Extruded Semiconductive Compound (Bonded or Strippable) to the insulation Metallic part: (by one of the following) • Copper tape is applied helically with a suitable overlap. • Copper Wires helically applied and banded with a Copper tape to achieve electrical contact • Overlapped Copper Tape then Copper Wires are also available (in single core only) Conductor screen, XLPE insulation and non metallic insulation screen are applied at the same time using triple head extruder.
Inner Sheath	Extruded PVC Compound
Armoring	Double Aluminum Tapes applied helically over the inner sheath.
Outer Sheath	Extruded PVC Compound
Outer Sheath Color	Black (default color), or any other color according to customer request
Maximum Operating Temp.	90° C
Minimum bending Radius	12 x outer diameter
Packing Condition	Our standard length on Wooden Drum / or according to customer request



Product Code	Nominal Cross Sectional Area	Max. Conductor Resistance			Operating Capacitance	Inductance		Current Rating				Approx. Overall Diameter	Approx. Weight
		DC at 20 °C	AC at 90° C			Flat	Trefoil	Laid in Ground		Laid in Free Air			
			Flat	Trefoil				Flat	Trefoil	Flat Touched	Trefoil		
													
mm²	Ω/km	Ω/km	Ω/km	μF/km	mH/km	mH/km	A	A	A	A	mm	kg/km	
A - Copper Conductors													
PM-CT14-X8-01-DP	50	0.3870	0.4937	0.4937	0.1836	0.5073	0.4610	224	217	245	241	30.4	1340
PM-CT15-X8-01-DP	70	0.2680	0.3420	0.3420	0.2087	0.4792	0.4330	276	266	303	301	32.2	1595
PM-CT16-X8-01-DP	95	0.1930	0.2465	0.2466	0.2294	0.4600	0.4138	328	315	366	365	33.9	1890
PM-CT17-X8-01-DP	120	0.1530	0.1957	0.1958	0.2486	0.4421	0.3959	372	360	424	420	35.3	2155
PM-CT18-X8-01-DP	150	0.1240	0.1588	0.1590	0.2692	0.4292	0.3830	410	402	481	474	37.0	2555
PM-CT19-X8-01-DP	185	0.0991	0.1273	0.1275	0.2937	0.4148	0.3686	461	454	551	545	38.8	2945
PM-CT20-X8-01-DP	240	0.0754	0.0974	0.0977	0.3429	0.4000	0.3538	530	528	649	642	41.5	3570
PM-CT21-X8-01-DP	300	0.0601	0.0783	0.0786	0.3547	0.3885	0.3423	588	588	742	735	43.9	4230
PM-CT22-X8-01-DP	400	0.0470	0.0621	0.0626	0.3912	0.3766	0.3304	657	658	844	838	46.8	5205
PM-CT23-X8-01-DP	500	0.0366	0.0496	0.0503	0.4357	0.3644	0.3182	735	734	973	954	50.3	6345
PM-CT24-X8-01-DP	630	0.0283	0.0398	0.0407	0.4909	0.3554	0.3091	810	829	1090	1097	54.6	7755
PM-CT25-X8-01-DP	800	0.0221	0.0328	0.0339	0.5500	0.3442	0.2980	884	912	1222	1234	59.2	9655
PM-CM26-X8-01-DP	1000	0.0176	0.0250	0.0253	0.6814	0.3138	0.2976	1062	1093	1507	1519	70.2	12775
B - Aluminum Conductors													
PM-AT14-X8-01-DP	50	0.6410	0.8219	0.8219	0.1892	0.5001	0.4539	173	169	191	188	30.8	1075
PM-AT15-X8-01-DP	70	0.4430	0.5682	0.5682	0.2101	0.4768	0.4305	212	207	236	234	32.3	1185
PM-AT16-X8-01-DP	95	0.3200	0.4106	0.4106	0.2322	0.4576	0.4114	254	247	287	284	34.1	1335
PM-AT17-X8-01-DP	120	0.2530	0.3247	0.3248	0.2500	0.4411	0.3949	288	281	332	326	35.4	1450
PM-AT18-X8-01-DP	150	0.2060	0.2645	0.2646	0.2733	0.4266	0.3804	320	315	375	370	37.3	1685
PM-AT19-X8-01-DP	185	0.1640	0.2109	0.2110	0.2978	0.4126	0.3664	361	358	432	428	39.1	1855
PM-AT20-X8-01-DP	240	0.1250	0.1611	0.1612	0.3249	0.4000	0.3538	417	414	509	503	41.5	2115
PM-AT21-X8-01-DP	300	0.1000	0.1292	0.1294	0.3547	0.3885	0.3423	461	465	585	577	43.9	2380
PM-AT22-X8-01-DP	400	0.0778	0.1011	0.1014	0.3912	0.3766	0.3304	528	530	679	673	46.8	2840
PM-AT23-X8-01-DP	500	0.0605	0.0794	0.0799	0.4357	0.3644	0.3182	596	602	779	776	50.3	3290
PM-AT24-X8-01-DP	630	0.0469	0.0625	0.0631	0.4922	0.3551	0.3088	672	683	908	900	54.7	2890
PM-AT25-X8-01-DP	800	0.0367	0.0502	0.0511	0.5500	0.3442	0.2980	751	765	1030	1032	59.2	4590
PM-AM26-X8-01-DP	1000	0.0291	0.0389	0.0391	0.6814	0.3138	0.2976	905	919	1277	1274	70.2	6145

The above data is approximate and subject to manufacturing tolerances

Note:-

Flame retardant PVC sheathed or Polyethylene Sheathed are available according to customer request
Longitudinal or radial Water tightness layers are also available according to customer request
the below data are applicable for cables with rated voltage 12.7/22 (24) KV